

1. INTRODUCTION

In recent years, the rapid expansion of social media platforms has transformed the way people communicate, interact, and express their emotions. People frequently share their thoughts, opinions, and feelings through online platforms such as WhatsApp, Instagram, Facebook, and Twitter.

Among these communications, positive and encouraging messages play an important role in promoting mental well-being and social harmony. Such positive and supportive content is commonly referred to as hope speech.

Hope speech refers to messages that express encouragement, optimism, emotional support, and motivation, especially toward individuals or communities facing challenges. The automatic detection of hope speech using computational techniques has recently gained importance as an emerging research area in the field of Natural Language Processing (NLP).

In multilingual regions such as coastal Karnataka, communication on social media often occurs in code-mixed language, where users combine Tulu and English within the same sentence. For example, messages like *“Don’t worry, yenkla time barpundu”* or *“Life tough aand, but hope undu”* contain both English and Tulu words. This mixture of languages increases the complexity of automated text analysis.

This project, titled **“Hope Speech Detection in Code-Mixed Tulu Language,”** aims to develop a machine learning-based system that can automatically classify whether a given code-mixed Tulu text contains hope speech or not.

The proposed system applies Natural Language Processing techniques, including text preprocessing, feature extraction, and classification algorithms, to analyze and categorize social media content effectively.

Tulu is considered a low-resource language, as it has limited annotated datasets and computational linguistic tools available for research. When Tulu is mixed with English in informal online conversations, variations in spelling, grammar, and the use of local slang further complicate the process of automatic text classification.

HOPE SPEECH DETECTION IN CODE-MIXED TULU LANGUAGE

The primary objective of this project is to contribute to regional language processing and promote positive digital communication. By accurately identifying hope speech, the system can support mental well-being initiatives, encourage constructive online interactions, and help reduce the spread of negative or harmful content.

In summary, this project demonstrates how Artificial Intelligence and Machine Learning can be effectively applied to analyze code mixed languages like Tulu and promote positivity in modern digital communication environment.

2. Related Work

With the rapid growth of social media platforms, large volumes of user-generated text are produced every day. This has led to increased research interest in Natural Language Processing (NLP) tasks such as sentiment analysis, emotion detection, hate speech identification, and offensive language classification. These studies aim to understand human emotions and behavior expressed through textual data.

Serval researches have focused on sentiment analysis, where text is classified as positive, negative, or neutral. Machine learning techniques such as Logistic Regression, Support Vector Machines (SVM), Naïve Bayes, and Decision Trees have been widely used to analyze sentiments in social media posts. These approaches have shown good performance for high-resource languages like English.

More recent research has shifted attention toward **hate speech and abusive language detection** to control the spread of harmful content online. While these studies help in reducing negativity, they mainly focus on identifying negative or offensive speech.

In contrast, **Hope speech detection** emphasizes identifying positive, encouraging, and supportive messages that promote optimism and mental well-being.

Hope speech detection is an emerging research area that aims to identify messages conveying encouragement emotional support, and motivation. Studies have shown that promoting hopeful and positive content can help counter online negativity and improve mental health outcomes. However, most existing work on hope speech detection is limited to widely spoken languages such as English and Hindi.

In multilingual regions, users often communicate using **code-mixed language**, where two or more languages are mixed within a single sentence. Processing code-mixed text is challenging due to irregular grammar, spelling variations, and lack of standardized linguistic resources Research on code-mixed language processing has mainly focused on English-Hindi or English-Tamil combinations

Very limited research has been carried out on low-resource regional languages such as Tulu. The absence of large annotated datasets and language-specific tools makes automated text classification difficult.

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When Tulu is mixed with English in social media communication, the complexity of language processing increases further. Based on the review of existing literature, it is evident that there is a significant research gap in **Hope speech detection for code-mixed Tulu language text**.

This project aims to address this gap by applying machine learning and NLP techniques to identify hope speech in code-mixed Tulu-English data, thereby contributing to regional language processing and promoting positive digital communication.

3. METHODOLOGY

The proposed Hope Speech Detection system follows a systematic and structured methodology to classify code-mixed Tulu-English text into hope speech and non-hope speech categories. The methodology consists of several stages, including data collection, preprocessing, feature extraction, model training, and evaluation.

3.1. Data Collection:

The dataset used in this project consists of code-mixed Tulu-English text collected from social media platforms such as WhatsApp groups, Instagram comments, and Facebook posts. The collected text data contains informal conversational messages where Tulu and English words are mixed within the same sentence.

The dataset is manually labeled into two classes:

- Hope Speech
- Non-Hope Speech

Manual annotation ensures that the classification reflects the presence or absence of encouraging, motivational, and supportive content.

3.2. Data Preprocessing

To Raw social media text often contains noise and inconsistencies. Therefore, preprocessing is performed before training the model. The following steps are applied:

Removal of URL and Tags

3.2.1. URL Removal

Social media texts frequently contain URLs or hyperlinks. These links do not contribute meaningful information for detecting hope speech.

3.2.2. Extra Whitespace Removal

Code-mixed text often includes special symbols such as “@”, “#”, “!”, emojis, and punctuation marks. While some symbols may carry emotional meaning, most do not contribute significantly to hope speech classification.

3.2.3. Lowercase Conversion

Text data may contain a mix of uppercase and lowercase letters. For example, “HOPE” and “hope” represent the same meaning but are treated differently by the model. To maintain uniformity, all text is converted to lowercase. This ensures consistency and prevents duplication of features.

3.2.4. Extra Whitespace Removal

Stopwords are common words such as “the”, “is”, “and”, “are”, which do not carry significant meaning in classification tasks. Removing stopwords reduces dimensionality and allows the model to focus on important words such as “hope”, “strong”, “don’t worry”, “success”, and similar expressions that indicate encouragement.

3.3. Feature Representation

After preprocessing, textual data must be converted into numerical form so that machine learning algorithms can process it.

3.3.1 TF-IDF (Term Frequency – Inverse Document Frequency)

TF-IDF is a statistical technique used to measure the importance of a word in a document relative to the entire dataset.

3.3.1.1. Term Frequency (TF)

Term Frequency measures how frequently a word appears in a sentence or document. Words that appear more often in hope-related sentences, such as “hope”, “strong”, “success”, and “don’t worry”, receive higher values.

3.3.1.2. Inverse Document Frequency (IDF)

Inverse Document Frequency reduces the importance of very common words that appear in many documents. For example, words like “is” or “and” appear in almost all texts and are not useful for classification.

3.3.2. Handling Code-Mixed Text in Feature Extraction

Since the dataset contains both Tulu and English words, feature extraction must handle:

- Mixed vocabulary
- Transliteration variations
- Informal grammar

3.4. Model Training

After completing feature extraction, the processed numerical data is used to train machine learning models. The purpose of model training is to enable the system to learn patterns that distinguish **hope speech** from **non-hope speech** in code-mixed Tulu-English text. The dataset is divided into training and testing sets. The training data is used to build the model, while the testing data is used to evaluate its performance on unseen samples. The following classification algorithms are implemented:

- **Logistic Regression:**

Is a supervised machine learning algorithm used for binary classification problems. It predicts the probability of an outcome using the sigmoid (logistic) function. It is widely used in text classification tasks like sentiment analysis and speech detection.

- **Support Vector Machine (SVM):**

Is a supervised machine learning algorithm used for classification tasks. In the training phase, SVM finds the optimal hyperplane that best separates different classes with maximum margin. It is effective for text classification problems like hope speech detection, especially with high-dimensional data.

- **Naïve Bayes:**

Is a supervised machine learning algorithm based on Bayes' Theorem and probability. It assumes that features are independent of each other (naïve assumption). It is widely used in text classification tasks like spam detection, sentiment analysis, and hope speech detection.

- **Random Forest:**

Is a supervised machine learning algorithm used for classification and regression. It works by building multiple decision trees during training and combining their outputs for better accuracy. It reduces overfitting and performs well for text classification tasks like hope speech detection.

3.5. Evaluation

After training the machine learning models, their performance is evaluated using the testing dataset. Evaluation is necessary to measure how effectively the models classify code-mixed Tulu-English text into hope speech and non-hope speech categories.

To assess the performance of the classifiers, the following evaluation metrics are used:

- **Accuracy:** measures the overall percentage of correctly classified instances.
- **Precision:** indicates how many texts predicted as hope speech are actually hope speech.
- **Recall:** measures how many actual hope speech instances are correctly identified by the model.
- **F1-Score:** provides a balanced measure of precision and recall.
- **Confusion matrix:** is used to visualize the classification results by showing true positives, true negatives, false positives, and false negatives.

4. EXPERIMENTS AND RESULTS

To evaluate the performance of the proposed Hope Speech Detection system, a series of experiments were conducted using the labeled code-mixed Tulu-English dataset. The experiments followed a structured pipeline including preprocessing, feature extraction, model training, and evaluation

4.1 Preprocessing

To build an effective Hope Speech Detection system for code-mixed Tulu language, it is essential to clean and prepare the raw textual data before feeding it into machine learning models. Social media text is often informal, unstructured, and noisy. Therefore, preprocessing is performed to improve data quality and enhance classification accuracy. The following steps were performed:

- **Removal of URLs:** Social media texts frequently contain hyperlinks and web addresses. These URLs do not contribute meaningful information for detecting hope speech.
- **Removal of Special Characters and Punctuations:**
Code-mixed text may contain special symbols such as “@”, “#”, “!”, emoji’s and other punctuation marks. While some symbols may express emotion, most are not useful for classification.
- **Removal of Extra Whitespace:** Social media text may include extra spaces, tabs, or line breaks. These unnecessary spaces can affect tokenization. Therefore, whitespace normalization is performed to ensure clean and properly formatted text.
- **Lowercase Conversion:** Text data may contain both uppercase and lowercase letters. For example, “HOPE” and “hope” represent the same meaning but are treated differently by machine learning models. To maintain consistency, all text is converted to lowercase.

4.2. Evaluation Metrics

To compare the performance of the models, standard evaluation metrics were used:

- Accuracy
- Precision
- Recall
- F1-Score

4.3. Final Prediction Output:

SVM Classification Results Table:

Model	Accuracy
SVM	0.6729 (~67%)

Detailed Classification Report (SVM):

Class	Precision	Recall	F1-Score	Support
Blended Hope	0.35	0.20	0.25	191
Discouraging Hope	0.59	0.23	0.33	153
Encouraging Hope	0.73	0.79	0.76	406
Uninvolved	0.69	0.88	0.78	534

Averages:

Metric Type	Precision	Recall	F1-Score	Support
Macro Avg	0.59	0.52	0.53	1284
Weighted Avg	0.64	0.67	0.64	1284

Random Forest Results Table:

Model	Accuracy
Random Forest	0.6596 (~66%)

Detailed Classification Report (Random Forest):

Class	Precision	Recall	F1-Score	Support
Blended Hope	0.32	0.11	0.16	191
Discouraging Hope	0.60	0.22	0.32	153
Encouraging Hope	0.72	0.76	0.74	406
Uninvolved	0.66	0.91	0.76	534

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Naïve Bayes Results:

Model	Accuracy
Naïve Bayes	0.6558 (~65.6%)

Classification Report (Naïve Bayes):

Class	Precision	Recall	F1-Score	Support
Blended Hope	0.55	0.09	0.15	191
Discouraging Hope	0.79	0.10	0.17	153
Encouraging Hope	0.63	0.86	0.73	406
Uninvolved	0.68	0.86	0.76	534

Voting Classifier Results

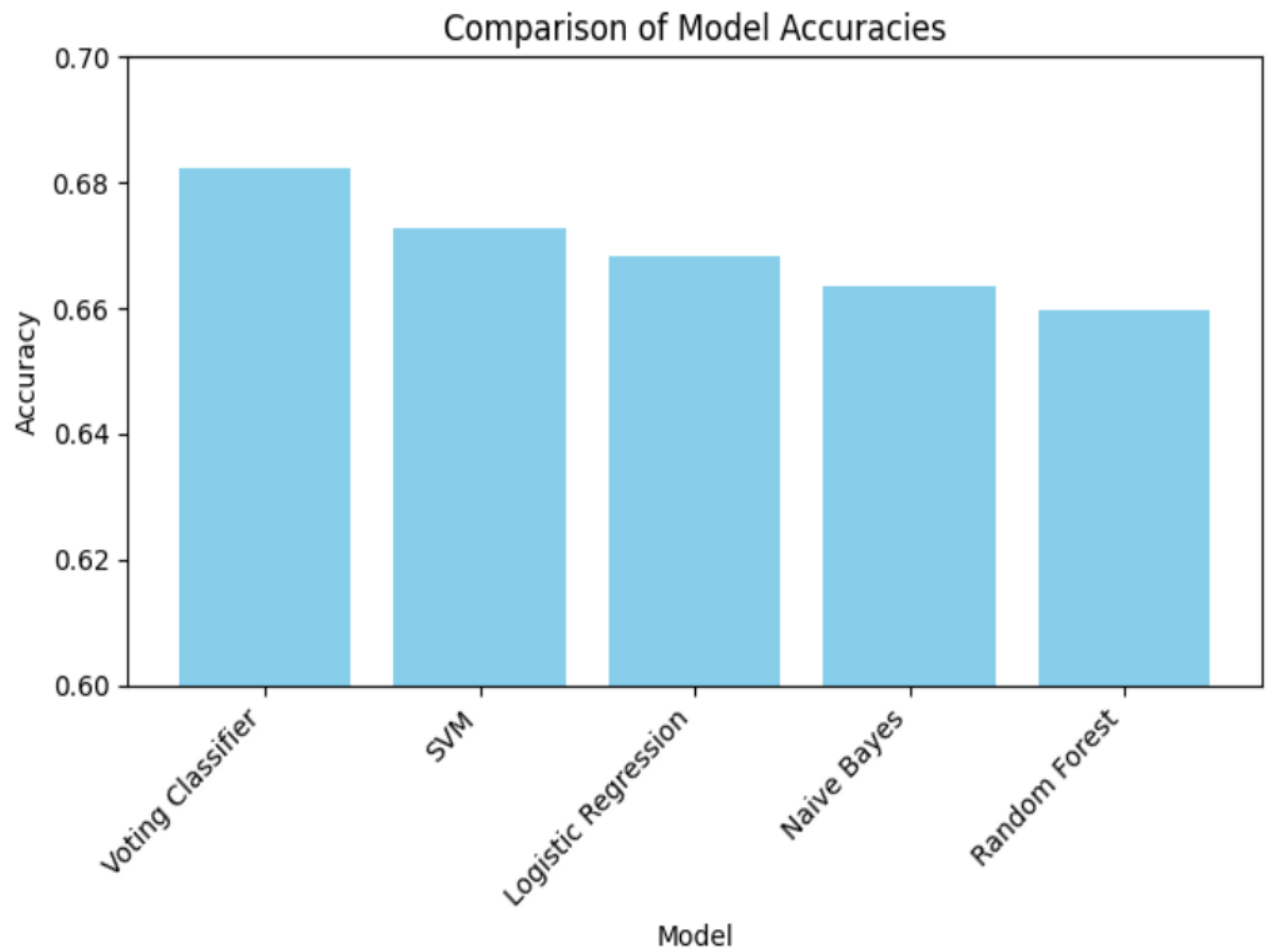
Model	Accuracy
Voting Classifier	0.6511 (~65.1%)

Classification Report:

Class	Precision	Recall	F1-Score	Support
Blended Hope	0.35	0.17	0.23	191
Discouraging Hope	0.46	0.23	0.31	153
Encouraging Hope	0.70	0.80	0.75	406
Uninvolved	0.68	0.83	0.75	534

Model Accuracy Comparison Table:

Model	Accuracy (%)	Performance Level
Support Vector Machine (SVM)	67.29%	High
Random Forest	65.96%	Moderate
Naïve Bayes	65.58%	Moderate
Voting Classifier	65.11%	High (Balanced)
Logistic Regression	63.38%	Moderate



5. CONCLUSION

The development of a Hope Speech Detection system for code-mixed Tulu language represents an important step toward promoting positive and inclusive communication in low-resource regional languages.

In multilingual communities, especially in regions where Tulu is widely spoken along with Kannada, English, and other languages, social media conversations often appear in a code-mixed format. This mixing of languages makes automatic text classification more challenging due to spelling variations, informal grammar, transliteration, and lack of standardized datasets.

In this project, we focused on identifying and classifying hope speech in code-mixed Tulu text. Hope speech plays a crucial role in encouraging harmony, reducing negativity, and fostering supportive digital environments. Unlike hate speech detection, which concentrates on identifying harmful content, hope speech detection emphasizes recognizing positive, motivational, and inclusive expressions.

By training and testing machine learning (or deep learning) models on annotated datasets, the system learns patterns that indicate supportive and uplifting language even in mixed language contexts.

One of the major contributions of this work is addressing the scarcity of computational resources for Tulu, which is considered a low-resource language. Handling code-mixed data required careful preprocessing steps such as text normalization, tokenization, removal of noise, and handling transliterated words.

Feature extraction techniques and classification algorithms were applied to improve prediction accuracy. The model performance was evaluated using standard metrics such as accuracy, precision, recall, and F1-score to ensure reliability.

The results demonstrate that automated systems can effectively identify hope speech even in complex code-mixed scenarios, although challenges remain due to limited data and linguistic diversity. Future improvements can include expanding the dataset, applying transformer-based models, incorporating sentiment analysis, and building real-time moderation tools.

Overall, this project contributes to the field of Natural Language Processing (NLP) by extending hope speech detection to a regional and code-mixed language setting. It promotes the creation of safer and more positive online spaces while supporting the technological development of underrepresented languages like Tulu.

With further research and larger datasets, the system can be enhanced to achieve higher accuracy and broader applicability across social media platforms.

6. REFERENCES

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